# System and Method for Cross-Layer Multi-Modal Threat Detection in Healthcare Environments

A Privacy-Preserving
Insider-Risk Platform for Critical Healthcare Infrastructure
Using Edge Sensor Fusion, Hashed Behavioral Monitoring,
Isolation-Forest Anomaly Detection, and Explainable Risk Scoring


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**TRL:** 4–5 (Lab-validated, simulated hospital environment)

# Problem Statement

## The Dual-Threat Challenge in Healthcare

Hospitals operate life-critical equipment (ventilators, infusion pumps, narcotics cabinets) facing **two simultaneous, uncorrelated threat classes**:

### Physical Threats

- Tampering with medical devices
- Unauthorized cabinet access
- Equipment theft or displacement
- Environmental faults (vibration, impact)

### Digital Threats

- Unauthorized host logins
- Malicious shell commands
- Privilege escalation attempts
- Insider policy violations

## Critical Constraint

Transmitting raw commands and patient-adjacent data violates **HIPAA**, **GDPR**, and **DPDP** regulations.

# Limitations of Existing Approaches

| Existing Technology | Limitation |
|---|---|
| Traditional EDR / Host IDS | Ships raw commands; no physical-world input; no healthcare context |
| Physical Security (CCTV, door sensors) | Operates in isolation from host behavior; no digital correlation |
| IoT Tamper Sensors | Local hardware alarm only; no integration with host analytics |
| Rule-based Insider Tools | Static ACLs; cannot detect legitimate credentials used in illegitimate *patterns* |
| Generic UEBA Products | Rely on raw log content; not hash-based; no sensor fusion |
| ML-based IDS | Opaque scoring; clinical staff cannot interpret or act on results |

## Key Gap Identified

**No existing solution** fuses physical + digital telemetry at the feature-vector level while preserving privacy and providing explainable, device-criticality-aware alerts.

# Proposed Invention

**A unified, cross-layer threat engine** that fuses physical and digital telemetry under a single anomaly model with privacy-by-construction.

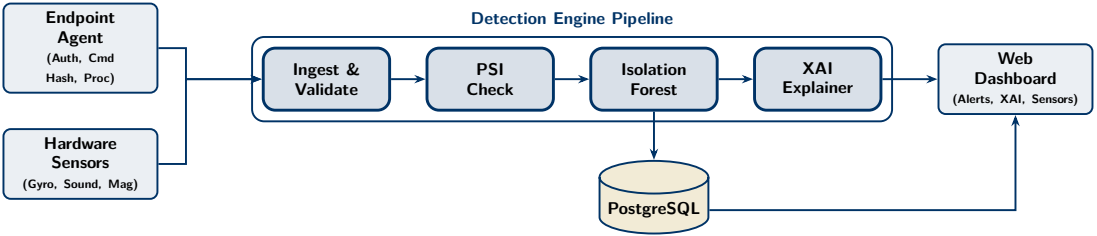
## Core Components

1. **Endpoint Agent** — Linux host monitor; transmits only SHA-256 command hashes
2. **Sensor Layer** — MPU-6050 (gyro/accel), HW-485 (sound), HW-509 (magnetic)
3. **Detection Engine** — PSI + Isolation Forest + XAI pipeline
4. **Web Dashboard** — Real-time alerts with explainable risk scores

## Key Differentiators

- Hash-only command telemetry (PSI)
- Cross-layer temporal fusion
- Device-criticality-weighted scoring
- Role/shift-aware insider baselining
- Clinician-readable XAI explanations
- Single engine replaces 3 vendor products

# System Architecture




- **Privacy enforced at schema level**: `command` events carry only `command_hash` — raw text never leaves the endpoint
- **Unified event schema**: All event types (`auth`, `command`, `process`, `sensor`) validated against a single JSON schema

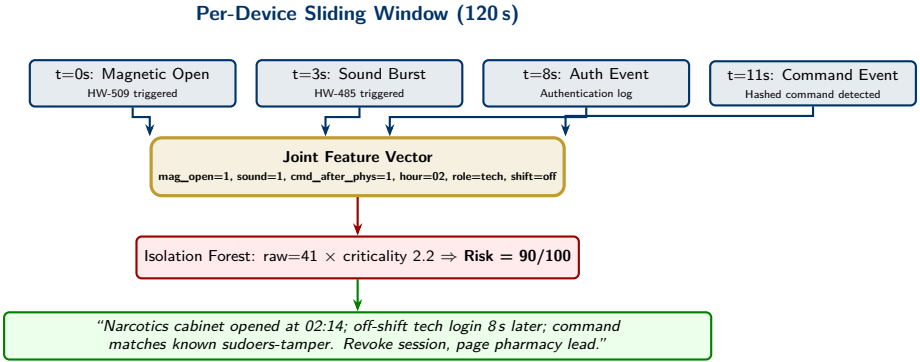
# Novelty and Inventive Step

## What Makes This Invention Non-Obvious?

No prior art combines all of these elements in a single system:

1. **Cross-Layer Temporal Fusion as a First-Class Feature**
   Physical + digital events fused at the *feature-vector level* before ML scoring, not merely concatenated at the UI level.
2. **Hash-Only Command Telemetry via PSI**
   Privacy enforced structurally by the schema and hash function — not by policy.
3. **Device-Criticality-Weighted Risk Scoring**
   Same anomaly scored differently on an ICU ventilator ($\times 2.0$) vs. a lobby kiosk ($\times 1.0$).
4. **Explainable AI for Non-ML Audiences**
   Clinician-readable paragraphs citing both physical and digital contributors.
5. **Role- and Shift-Aware Insider Baselining**
   Unsupervised behavioral envelopes per staff role — detects pattern deviations, not identity.
6. **Single Engine, Single Model Surface**
   Covers outsider attacks, insider misuse, physical tampering, and device-health precursors.

# Key Innovation: Cross-Layer Temporal Fusion




- A **pure EDR** sees only a low-severity suspicious command
- A **pure physical alarm** sees only a routine cabinet opening
- **Only the fused engine** produces the correct **high-severity, actionable alert**

# Proposed Patent Claims

## Independent Claims

**Claim 1.** **System Claim** — A healthcare threat-detection system comprising a Linux endpoint agent transmitting only normalized SHA-256 command hashes, physical sensors (MEMS inertial + acoustic + magnetic), and a detection engine with unified event schema.

**Claim 2.** **Method Claim** — A method of detecting cross-layer threats by: (i) collecting hashed behavioral events, (ii) collecting physical sensor events, (iii) fusing within a temporal window, (iv) scoring with device-criticality-weighted Isolation Forest, (v) producing human-readable explanation.

**Claim 3.** **CRM Claim** — A non-transitory computer-readable medium storing instructions to perform the method of Claim 2.

## Key Dependent Claims

**Claim 4.** Per-device temporal fusion window with joint feature vectors

**Claim 5.** PSI-style set-membership test over SHA-256 hashes

**Claim 6.** XAI module generating natural-language justifications

**Claim 7.** Role/shift-aware insider-risk baselining

**Claim 8.** Federated multi-hospital deployment sharing only model parameters

# Advantages and Market Potential

## Advantages Over Prior Art

- **Privacy by construction** — HIPAA/GDPR/DPDP compliant; raw commands never leave endpoint
- **Lower false-positive rate** — cross-layer context suppresses benign-looking anomalies
- **Higher detection of subtle attacks** — catches multi-step physical → digital chains
- **Criticality-aware triage** — ICU ventilator alerts surface above lobby printer alerts
- **Actionable XAI** — reduces mean time to action without needing a data scientist
- **Low cost** — commodity sensors ($5–15 each), open-source software stack
- **Replaces 3 vendor products** with one unified platform

## Healthcare Cybersecurity Market

| Year | Market Size |
|---|---|
| 2024 | ~USD 20 B |
| 2026 | ~USD 27 B |
| 2028 | ~USD 38 B |
| 2030 | ~USD 52 B |

**Growth drivers:**

- Hospital ransomware surge
- FDA pre-market cyber requirements
- EU MDR, HIPAA enforcement
- Hospital IoT proliferation
- AI-driven security adoption

# Summary and Conclusion

## Invention Summary

The proposed system is the first cross-layer, privacy-preserving, explainable threat-detection platform purpose-built for healthcare infrastructure.

### Patentability Criteria Met

- ✓ **Novelty** — No prior art combines hash-only PSI + sensor fusion + Isolation Forest + XAI + insider baselining
- ✓ **Inventive Step** — Cross-domain insight required (cybersecurity + IoT + ML + clinical workflow)
- ✓ **Industrial Applicability** — Validated at TRL 4–5; commodity hardware; clear path to deployment

### Next Steps

- Clinical pilot in an operational hospital ward (TRL 6)
- Independent red-team tamper exercises
- Quantitative evaluation: expected **+18–25% F1** improvement from fusion
- Federated multi-hospital validation
- Filing of provisional patent application